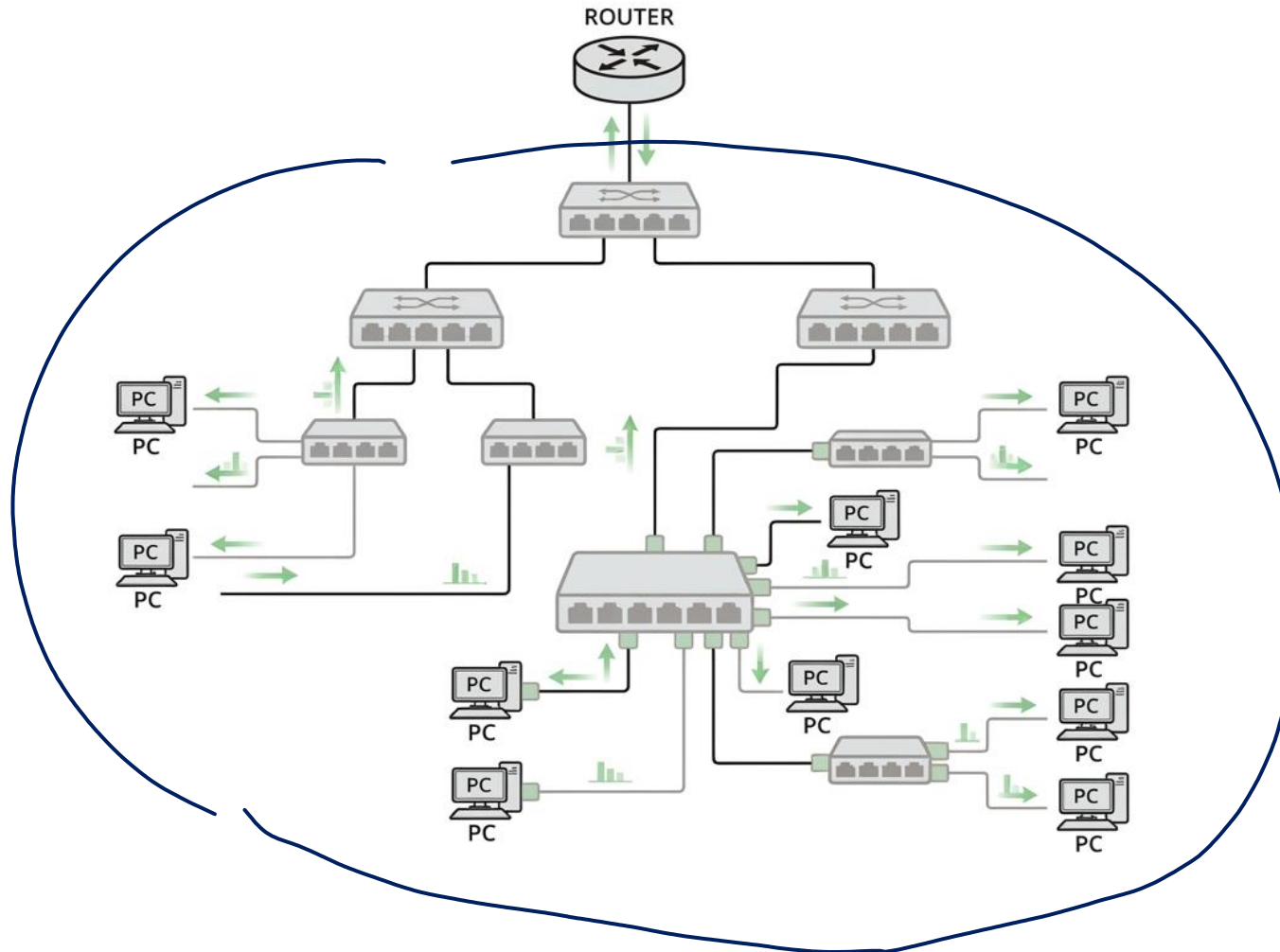
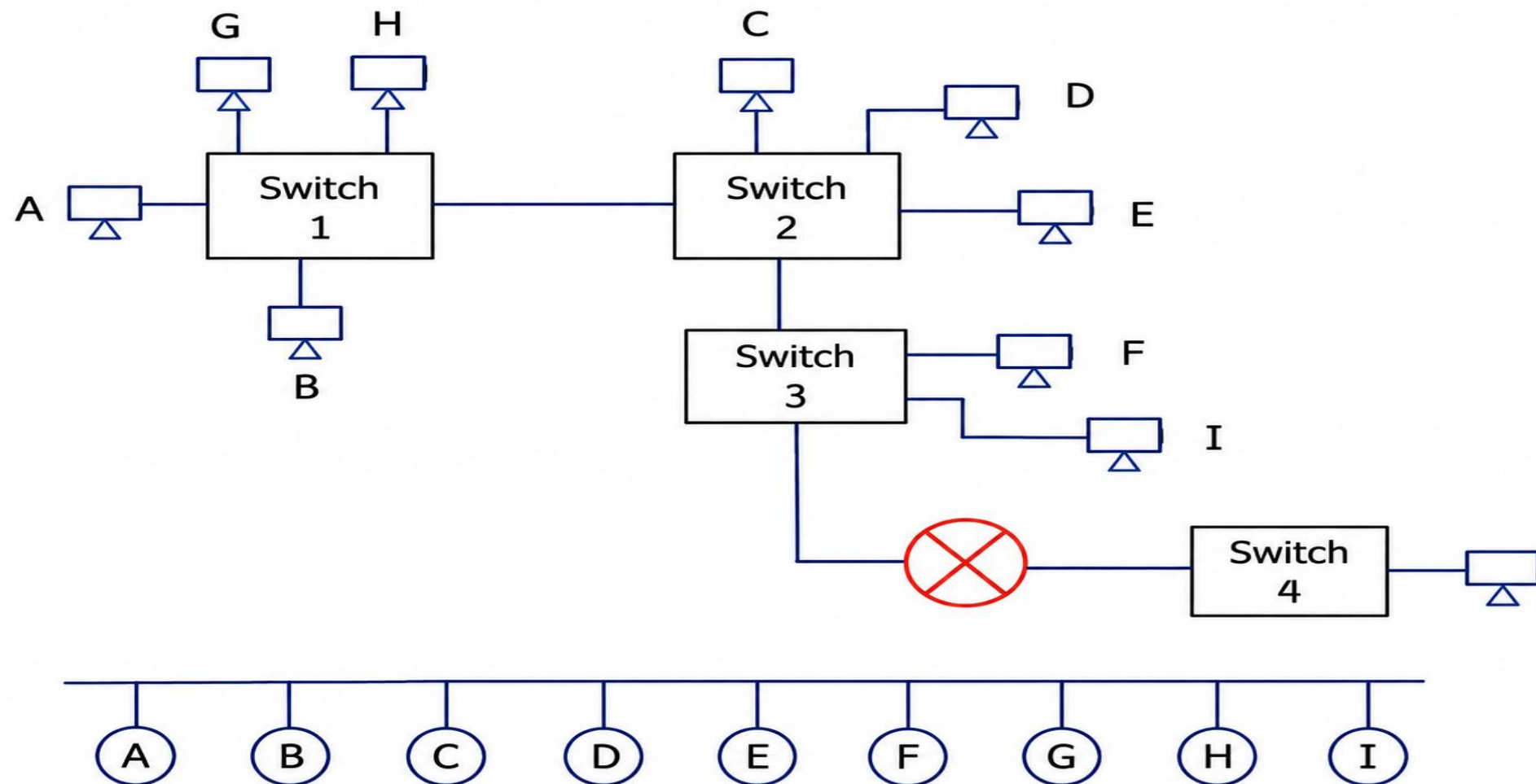


Data Link Layer Attack

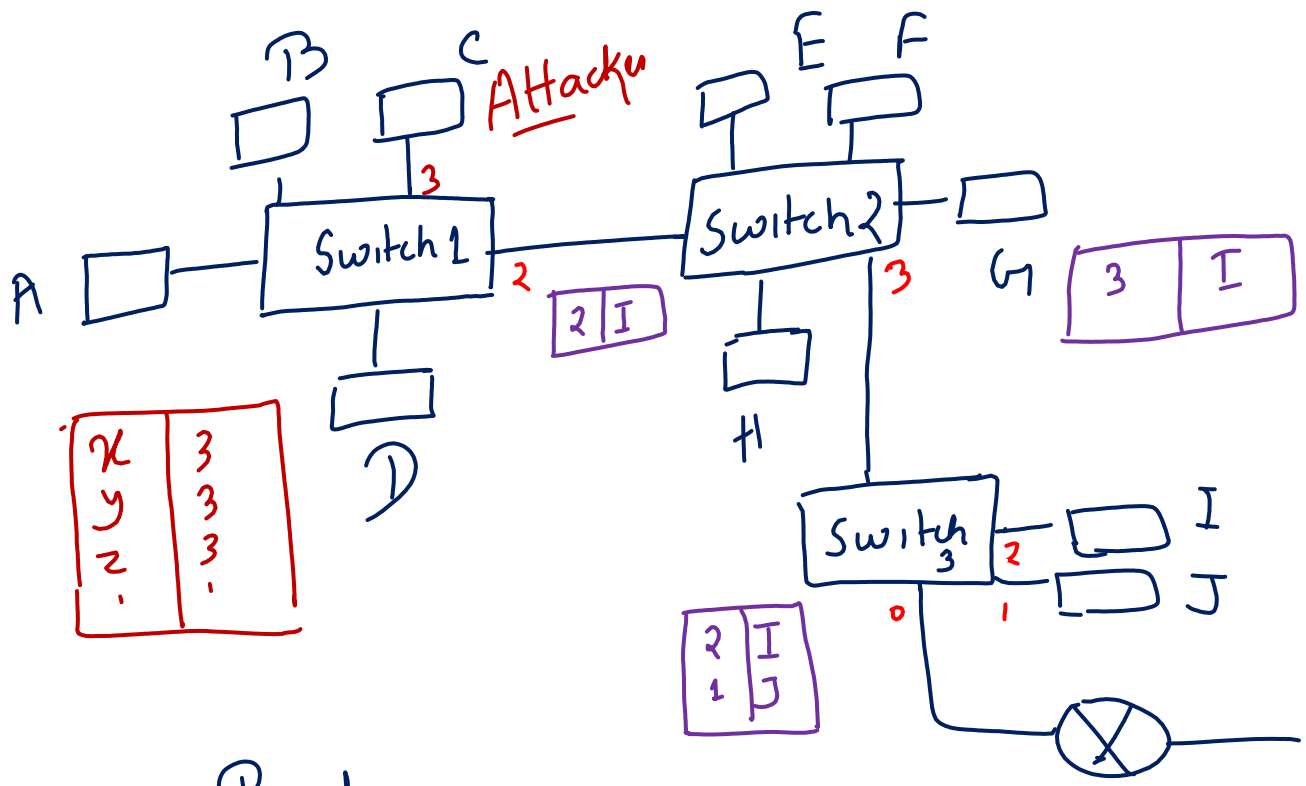
-Dr. Sumana Maiti

Data Link layer





These are neighbours.

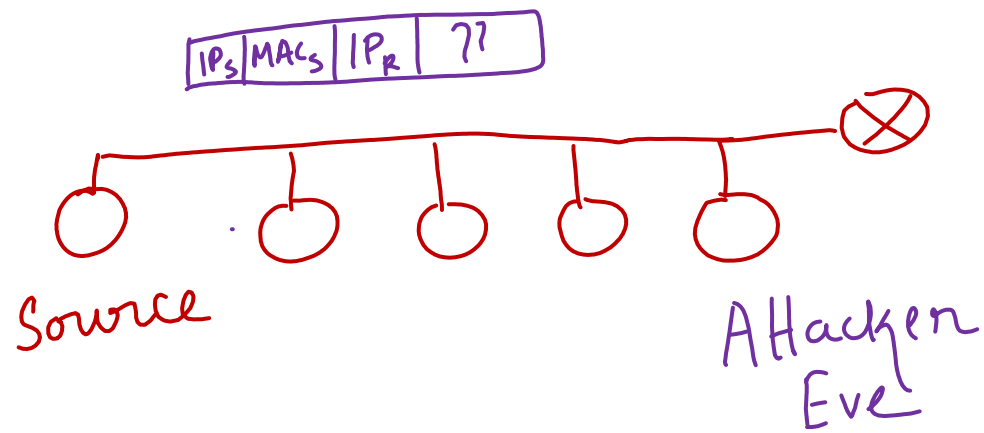


Learning Bridge

A → J

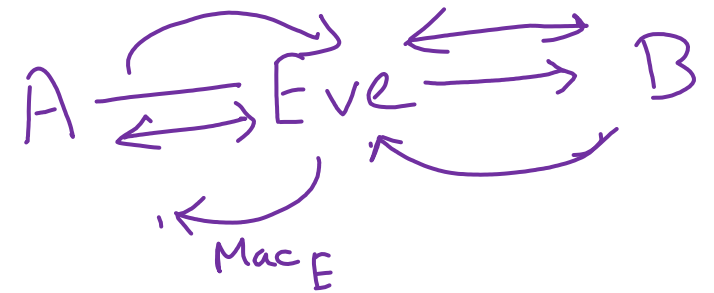
A → I

https



ARP is stateless

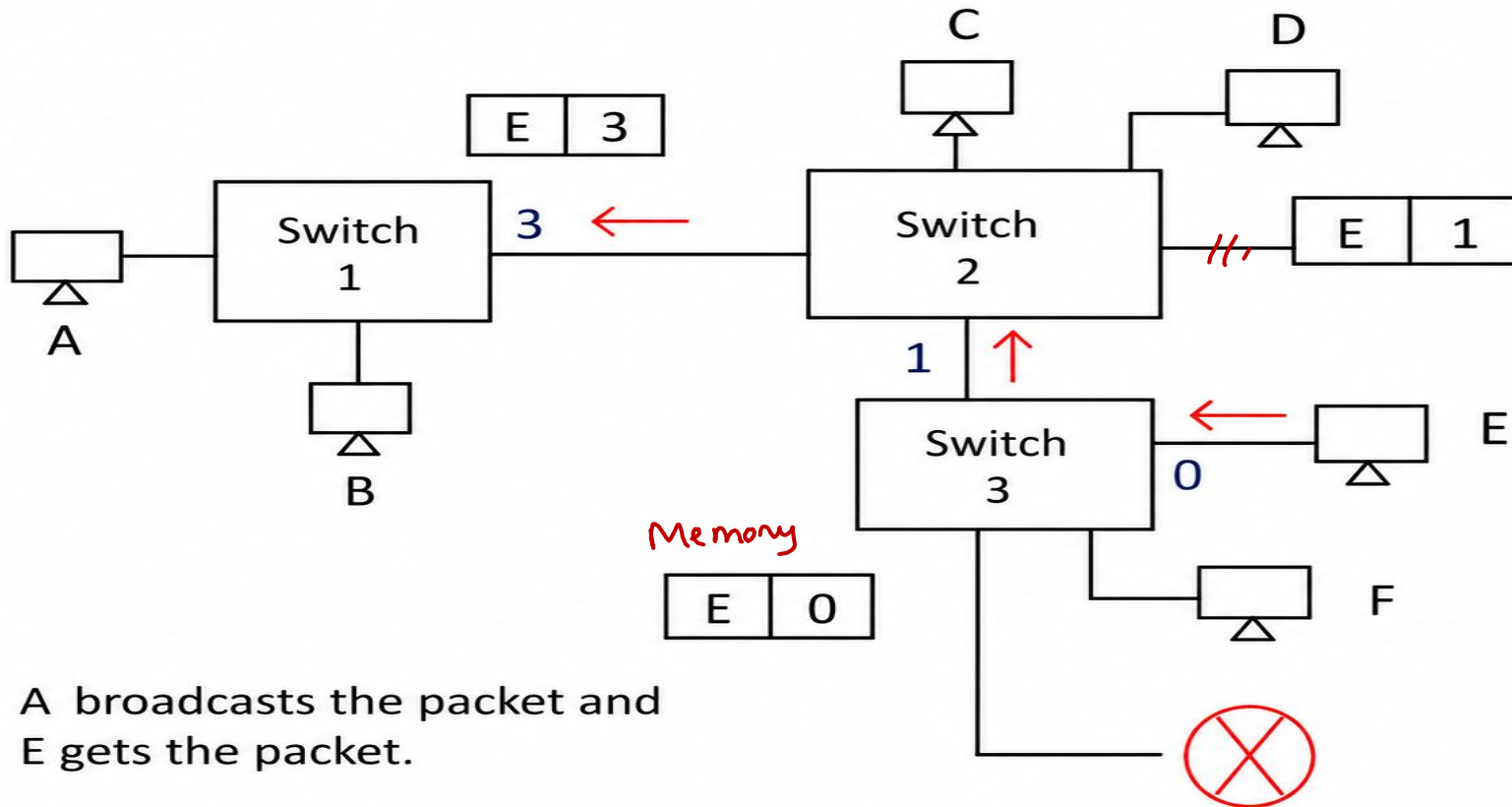
IP _R	MAC _E
-----------------	------------------



ARP spoofing
can lead to
Man in the
middle attack

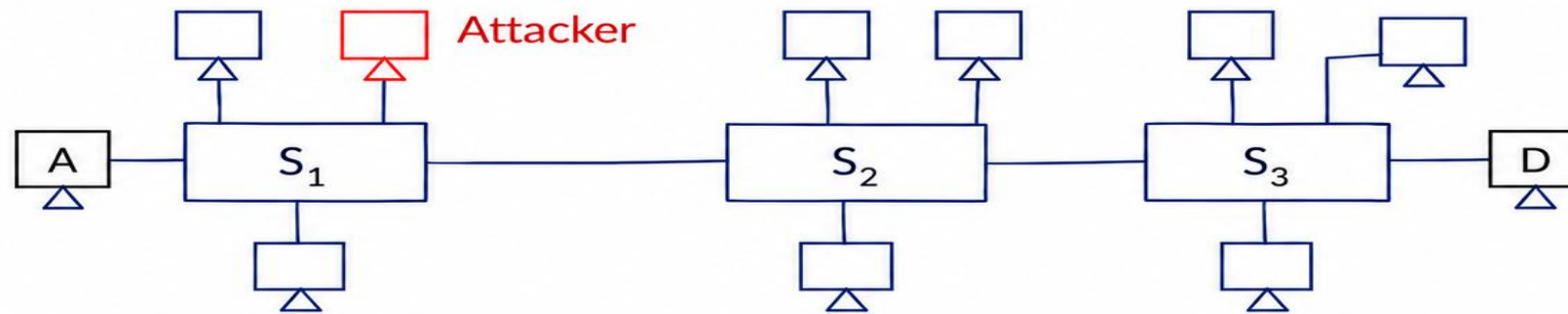
Eve is spoofing
the Router's Mac
address

Learning Bridge



- ① A broadcasts the packet and E gets the packet.
- ② When E sends the ack, each switch get the packet and store in a table from which port s/he gets the packet.

MAC Flooding



①

1	C
2	D
3	A
⋮	⋮

Each switch maintains a fixed amount of memory.
The switch is learning and storing the path in memory.

②

If the switch receives a packet (eg: destination x), but there is no entry for x in the table, the switch will broadcast to all ports.

③

Attacker keeps sending different packets (from different sources). No more space is left in memory for correct address.

④

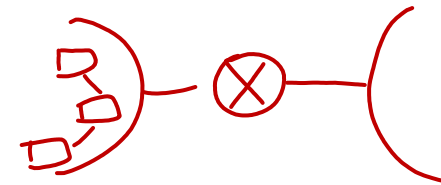
From next time, switch will broadcasts the packet.

What is defense mechanism?

- *Learn only one MAC address from one port. Will it work?*
- One switch has multiple hosts.
- Check whether the port is connected to another switch or end user.
- If the port is connected to end user, do MAC limiting.
- If we use https, it inherently provides confidentiality. If attacker gets the packet, s/he cannot see what is inside.

ARP Spoofing

- www.iitbbs.ac.in —> 10.10.40.191
- In network layer, source ip and destination ip will be added.
- Check whether the destination IP is part of same network or not (using subnet mask)
- **If the destination IP is within same network**, then to solve MAC address, source sends ARP request.
 - Source IP: IP_s ,
 - Source MAC: MAC_A
 - Destination IP: 10.10.40.191
 - Destination MAC: ??
- IIT BBS server replies the MAC address (ARP reply) and source stores it in ARP cache



Man-in-the middle attack

ARP Spoofing (Continue)

- **If the destination IP is not a part of source network**, it is the responsibility of router to take care of the packet.
- In this case, the ARP request is sent to know the MAC address of router.
 - *Source IP: IP_s ,*
 - *Source MAC: MAC_A*
 - *Destination IP: 10.10.40.191*
 - *Destination MAC: ??*
- ARP is stateless protocol, whenever the reply comes, it puts in **ARP cache**.
- Attacker can reply with router's IP with his own MAC address to source and source will store it.
- Here, **the attacker is spoofing the router.**

PHASE 1: SPOOFING ATTACK INITIATION

STEP 1: Attacker sends malicious ARP replies to Client A and Server B.



PHASE 2: TRAFFIC INTERCEPTION AND MAN-IN-THE-MIDDLE



STEP 2: Traffic is redirected through Attacker M, enabling eavesdropping, data modification, and traffic forwarding.

ARP CACHE STATUS (BEFORE vs. AFTER)

Cache	MAC	Cache	After
Poisoned	IP B is MAC M	Poisoned	IP B is MAC B
...	...	Poisoned	IP B is MAC M



Eavesdropping
(Read Private Data)

ATTACK IMPACT

Data Modification
(Change Traffic)



Session Hijacking
(Steal Login Info)

How to prevent this attack?

Switches need to observe which is odd and raise a flag.